

Name-Free Combinators for the Rho Calculus

A repair, a linear encoding, and the one operation that was missing

F1R3FLY.io research note

September 2026 — draft 3 (rewritten)

Abstract

The name-free concurrent combinators of *Name-free combinators for concurrency* were proposed as a target into which the rho calculus embeds with no binders at all. This note asks whether the embedding is correct, and finds three things. The combinator calculus as published is unsound, and the defect is one equation: the structural identification $*@P \equiv P$, which makes every running process a redex. It is removed, and the identification recovered where it is wanted through the semantic substitution of the original Meredith–Radestock presentation — whose drop clause turns out to be the combinator rewrite for $\mathbf{e}$, promoted from the metatheory to a reduction rule. Second, with the defect repaired the combinators still fall short of the rho calculus, and by exactly one capability: they have a name destructor and no name constructor, so they can open a name that arrives at runtime but never build one. A no-name-creation lemma turns this into a separation theorem. A small family of constructors closes the gap — four suffice, each building the quotation of one node from names, chief among them $\text{cons}_|(a, b, c)$ for parallel composition — and the translation then covers the whole calculus with no fragment restriction. Third, reflection buys an unexpected efficiency: a continuation can be quoted and released by a gate rather than rewritten atom by atom, so prefix elimination is linear where Yoshida’s is $\Theta(3^d)$ in the nesting depth of inputs. Correctness is established as full abstraction relative to encoded contexts, in the sense of the companion note, and the restriction to encoded observers is shown to be forced. A final section gives an alternative presentation in which $\mathbf{m}$ and $\mathbf{q}$ are not primitive but the nullary members of the constructor family, leaving a calculus of routers, one destructor, and constructors.

Contents

1	Introduction	2
2	The rho calculus	3
3	The rho combinators	4
3.1	Atoms and rules	4
3.2	The defect	5
3.3	The storage combinator	5
3.4	The name constructors	6
4	Why the original presentation is the right one here	6
5	Context-labelled bisimilarity	7

6	The translation	7
6.1	Seeds	7
6.2	Gate and distributor	8
6.3	Clauses	8
7	Correctness	9
8	Complexity of the target term	12
9	What reflection needs	12
9.1	The combinators cannot compute names	12
9.2	Minimality	13
10	An alternative presentation: constructors only	13
11	Corrigenda to [5]	14
12	Open questions	15

1 Introduction

The rho calculus¹ has no name-restriction operator: names are quoted processes, and freshness is obtained by quotation rather than by a binder. It nonetheless has a binder, the input prefix. Yoshida showed how to eliminate the input binder of the asynchronous π -calculus in favour of combinators, at the cost of keeping **new**. The combination is suggestive: a calculus with no **new**, compiled by a discipline that removes the remaining binder, should give a concurrency calculus with no bound names whatsoever. That is what [5] proposed. This note asks whether the proposal works.

Findings.

1. **One equation breaks the calculus.** The published combinators inherit $*@P \equiv P$ from a presentation in which quote and drop are mutually inverse up to structural congruence. In the combinator setting the drop is not a process former but a guard, and since every name is $@Q$ for some Q , the identification makes an arbitrary running process match the guard. Proposition 3.4: for all P, Q , $Q \mid \mathbf{m}(@Q, @P) \longrightarrow P$. The repair is to drop the equation, retain $@*x \equiv_{\mathcal{N}} x$, and let semantic substitution do the rest (§4).
2. **Reflection was half-implemented.** The rule $\mathbf{e}(a) \mid \mathbf{m}(a, @P) \longrightarrow P$ opens a name that arrives at runtime. Nothing builds one. Lemma 9.1: combinator reduction never produces a name that was not already present in the redex, hereditarily. Rho reduction does, at every step, because substituting under a quotation turns $@(*y \mid R)$ into $@(Q \mid R)$ — a name in neither participant. Theorem 9.2 converts this into a separation: no combinator term of the published calculus is equivalent to $\mathbf{for}(y \leftarrow a) @(*y \mid *y)!(\mathbf{0})$.

¹The name is an acronym and a pun. It abbreviates the **reflective higher order** calculus; dropping the periods from r.h.o. gives the English transliteration of the Greek letter following π , marking the calculus as successor to the π -calculus. It is written “rho calculus”, never with the Greek letter.

3. **Four constructors close the gap.** The missing operation is the construction of a name from names. It is supplied by constructors of the shape $\text{cons}_|(a, b, c) \mid \mathbf{m}(a, @P) \mid \mathbf{m}(b, @Q) \longrightarrow \mathbf{m}(c, @(P \mid Q))$, one for each node a rebuilt name can have. Proposition 3.8 shows four are enough for the whole rho calculus: parallel composition, message, and the two atoms that can head a translated input. There is no name-opaque fragment to restrict to.
4. **The storage combinator is about opacity, not power.** $\mathbf{q}(a, P)$, released by a forwarder rather than by a message, satisfies the no-name-creation lemma and so adds nothing to expressiveness. What it adds is silence: it is a message that only a redirection can read, which is what stored code and intermediate results of name construction need if the encoding is to be opaque to observers (§3.3, Remark 7.12).
5. **Prefix elimination is linear.** Reflection allows a continuation to be quoted once and released by a three-atom gate. Yoshida must instead push the prefix through the body, splitting at every parallel composition and delaying every atom, which costs a constant factor per level of nesting: Proposition 8.2 computes $(3^{d-1} + 1)/2$ atoms on an explicit family where the reflective encoding gives $3d + 1$.
6. **Correctness is full abstraction relative to encoded contexts.** Following [4], observation is by context-labelled bisimilarity with the observer class taken to be the image of the translation on source contexts. Theorem 7.9; and Proposition 7.10, that the restriction is forced rather than convenient, by the same argument that forces it there.

§10 gives the alternative presentation in which $\mathbf{m}$ and $\mathbf{q}$ are the nullary constructors. §11 lists the corrigenda to [5].

Notation. [5] writes $*$ for three distinct things: Yoshida’s replication, the rho calculus drop, and the combinator that consumes a message and runs its payload. We write $\mathbf{e}$ for the last and reserve $*x$ for the drop. The distinction between the second and the third is one of the points of §4.

2 The rho calculus

We follow [3].

Definition 2.1 (Syntax).

$$\begin{aligned} P, Q &::= \mathbf{0} \mid \text{for}(y \leftarrow x)P \mid x!(P) \mid *x \mid P \mid Q \\ x, y &::= @P \end{aligned}$$

$\text{fn}(P)$ is as usual, with $\text{fn}(\text{for}(y \leftarrow x)P) = \{x\} \cup (\text{fn}(P) \setminus \{y\})$. $\mathcal{N}^*(P)$ is the set of names occurring anywhere in P , including hereditarily inside quotations.

Definition 2.2 (Structural congruence, name equivalence). $\equiv$ is the least congruence containing α -equivalence and making $(\text{Proc}, |, \mathbf{0})$ an abelian monoid. $\equiv_{\mathcal{N}}$ is the least congruence on names with $@*x \equiv_{\mathcal{N}} x$, and $@P \equiv_{\mathcal{N}} @Q$ whenever $P \equiv Q$.

The converse identification $*@P \equiv P$ is deliberately absent. Its work is done by substitution.

Definition 2.3 (Semantic substitution). On names, $y\{\textcircled{Q}/y\} = \textcircled{Q}$ and $\textcircled{P}\{\textcircled{Q}/y\} = \textcircled{(P\{\textcircled{Q}/y\})}$. On processes, the structural clauses together with

$$(*y)\{\textcircled{Q}/y\} = Q, \quad (*z)\{\textcircled{Q}/y\} = *z \quad (z \neq y).$$

The drop clause is the whole of the difference between semantic and syntactic substitution. Syntactic substitution returns $*\textcircled{Q}$, and one then needs $*\textcircled{Q} \equiv Q$ to recover the meaning. Semantic substitution decodes at the moment of substitution and returns a process bisimilar to any other faithful decoding rather than congruent to a canonical one; the decoder is free to return an optimised representative.

Definition 2.4 (Reduction).

$$\frac{x_t \equiv_{\mathcal{N}} x_s}{\text{for}(y \leftarrow x_t)P \mid x_s!(Q) \longrightarrow P\{\textcircled{Q}/y\}} \quad \frac{P \longrightarrow P'}{P \mid Q \longrightarrow P' \mid Q} \quad \frac{P \equiv P' \quad P' \longrightarrow Q' \quad Q' \equiv Q}{P \longrightarrow Q}$$

Definition 2.5. P is *guarded-drop* if every $*x$ in P has x a variable bound by an enclosing input.

Lemma 2.6. *Guarded-drop is preserved by reduction, and no subterm $*\textcircled{Q}$ ever arises from a guarded-drop term.*

Proof. A drop can acquire a ground name only by substitution, and Definition 2.3 eliminates the drop rather than instantiating it. Every other clause is structural. $\square$

A term $*\textcircled{Q}$ may still be written by hand, and is then inert. For such terms $*\textcircled{Q}$ and Q are related by the decoding bisimulation rather than by $\equiv$, which is the latitude the original presentation was designed to leave. Source terms are assumed guarded-drop throughout.

3 The rho combinators

3.1 Atoms and rules

Definition 3.1 (Syntax).

$$\begin{aligned} A ::= & \text{m}(a, b) \mid \text{d}(a, b, c) \mid \text{k}(a) \mid \text{fw}(a, b) \mid \text{b}_l(a, b) \mid \text{b}_r(a, b) \mid \text{s}(a, b, c) \\ & \mid \text{e}(a) \mid \text{q}(a, P) \\ & \mid \text{cons}_l(a, b, c) \mid \text{cons}_m(a, b, c) \mid \text{cons}_d(a, b, c, e) \mid \text{cons}_s(a, b, c, e) \\ P, Q ::= & \mathbf{0} \mid A \mid P \mid Q \quad a, b, c ::= \textcircled{P} \end{aligned}$$

The first seven are Yoshida's, with her readings: d duplicates a message to two locations, k discards it, fw forwards it, b_r and b_l bind the received name for output and for input respectively, and s synchronises without passing a value. Note that $\text{m}(a, b)$ takes two *names*; since every name is a quoted process, $\text{m}(a, \textcircled{P})$ is the same atom written with its payload displayed. The last three are new and are discussed below.

Definition 3.2 (Structural congruence, name equivalence). $\equiv$ is the least congruence making $(\text{Proc}, \mid, \mathbf{0})$ an abelian monoid. There is no clause for α -equivalence, there being no binders, and *no clause identifying $*\textcircled{P}$ with P* . Name equivalence $\equiv_{\mathcal{N}}$ is as in §2.

Definition 3.3 (Reduction).

$$\begin{array}{ll}
d(a, b, c) \mid m(a, v) \longrightarrow m(b, v) \mid m(c, v) & b_r(a, b) \mid m(a, v) \longrightarrow fw(b, v) \\
k(a) \mid m(a, v) \longrightarrow \mathbf{0} & b_l(a, b) \mid m(a, v) \longrightarrow fw(v, b) \\
fw(a, b) \mid m(a, v) \longrightarrow m(b, v) & s(a, b, c) \mid m(a, v) \longrightarrow fw(b, c) \\
e(a) \mid m(a, @P) \longrightarrow P & fw(a, b) \mid q(a, P) \longrightarrow m(b, @P) \\
cons_l(a, b, c) \mid m(a, @P) \mid m(b, @Q) \longrightarrow m(c, @(P \mid Q)) & \\
cons_m(a, b, c) \mid m(a, u) \mid m(b, v) \longrightarrow m(c, @m(u, v)) & \\
cons_d(a, b, c, e) \mid m(a, u) \mid m(b, v) \mid m(c, w) \longrightarrow m(e, @d(u, v, w)) & \\
cons_s(a, b, c, e) \mid m(a, u) \mid m(b, v) \mid m(c, w) \longrightarrow m(e, @s(u, v, w)) &
\end{array}$$

closed under parallel composition and under $\equiv$, with subjects matched up to $\equiv_{\mathcal{N}}$.

Reduction is nondeterministic and is meant to be. Two consumers may compete for one message, as in $d(a, b, c) \mid k(a) \mid m(a, v)$; and two producers may compete for one consumer, as in $fw(a, b) \mid m(a, v) \mid q(a, P)$, where fw is the consumer at a and m and q both offer a value there. Neither is pathological and neither is new: they are the two sides of the race the communication rule has always had.

3.2 The defect

Proposition 3.4. *Suppose Definition 3.2 is extended with $*@P \equiv P$, with e written $*$ so that the drop and the drop combinator are one syntactic form. Then for all P, Q ,*

$$Q \mid m(@Q, @P) \longrightarrow P.$$

Proof. Every name is $@Q$ for some Q , so $*(@Q) \equiv Q$ by the added equation. Reduction is closed under $\equiv$, so the redex $*(a) \mid m(a, @P)$ with $a = @Q$ matches $Q \mid m(@Q, @P)$, and the rule fires. $\square$

An arbitrary running process can therefore be annihilated and replaced by an arbitrary payload, by anyone who knows its quotation. No faithfulness theorem survives this.

Remark 3.5. The repair is to omit the equation and to recognise $e(a)$ as an *atom* rather than a process former: it is the combinator corresponding to $\text{for}(y \leftarrow a)*y$, a guard that listens on a and runs what arrives. [5] lists it under the process production, alongside parallel composition, which is what invites the identification with the drop. Nothing else in [5] depends on the discarded equation: it occurs exactly once, in the definition of the structural congruence, while every other appearance of a drop there stands in payload position inside a quotation and requires $@*x \equiv_{\mathcal{N}} x$, the half that is kept.

3.3 The storage combinator

Remark 3.6 (q is a message only a forwarder can read). The rule for q has the same right-hand side as the rule for fw . So $q(a, P)$ and $m(a, @P)$ are indistinguishable to a forwarder, and differ everywhere else: a message at a may be consumed by any of d , k , b_r , b_l , s , e , a constructor or fw , whereas $q(a, P)$ may be consumed only by fw . Since $fw(a, a) \mid q(a, P) \longrightarrow m(a, @P)$, q is not a message awaiting a trigger but the introduction form for a name constant, of which m is the already-manifest case. Writing the payload as a name, $q(a, v)$ with $fw(a, b) \mid q(a, v) \longrightarrow m(b, v)$, makes the parallel with $m(a, v)$ exact.

$\mathbf{q}$ adds no expressive power — it satisfies Lemma 9.1 — and is not needed for the translation to be definable. What it provides is opacity, and the need for it is best stated after the constructors are in hand; see Remark 7.12.

3.4 The name constructors

Every rule but the last four moves names around or opens one. The last four build one.

Remark 3.7. Each constructor takes names and returns a name: it assembles the quotation of a single node from the quotations of its children, and the assembled name is delivered as an ordinary message, so it may then be used as a subject, a payload, or the input to a further constructor. Chaining them builds a name of any shape from the outside in. The operation so performed is substitution of a received name into a static term under a quotation, which is precisely what the rho communication rule does and what [5] has no counterpart for.

Proposition 3.8. *The four constructors of Definition 3.3 suffice for the translation of Definition 6.6: every name that must be rebuilt at runtime is assembled by a chain of them.*

Proof. A name to be rebuilt is $@[S]$ for a source term S with the bound name free in it, and $[-]$ is a homomorphism, so the nodes of $[S]$ are those of S . Take the path from an occurrence to the root; every node off the path is static and is supplied as a constant message, joined on by cons_i . On the path, by Definition 6.6: an occurrence under $*-$ is the received name itself, needing no constructor; a parallel composition needs cons_i ; an output $v!(T)$ has image $\mathbf{m}([v], @[T])$ and needs cons_m , which covers both a dynamic subject and a dynamic payload; and an input $\text{for}(z \leftarrow v)T$ has image $D([v]; \vec{p}) \mid G \mid \vec{L}$ in which $[v]$ occurs only in the head atom, which is $\mathbf{d}([v], p_0, q_1)$ when z occurs in T and $\mathbf{s}([v], b, c)$ when it does not; the remainder is static. Hence cons_d and cons_s . Duplication of the received name, when an occurrence appears at several places, is supplied by $\mathbf{d}$. $\square$

Remark 3.9 (The optimisations are not free). Proposition 3.8 is stated for the general clauses. The shortcuts of Remark 6.7 put a bound name into the argument of an $\mathbf{e}$ or a $\mathbf{fw}$, so a translation that takes them needs cons_e and cons_{fw} as well. Four is the count for the uniform translation, not a lower bound over all translations.

Remark 3.10 (A more exotic alternative). The family can be collapsed into one atom indexed by a static context: $\text{cons}[C](a, c) \mid \mathbf{m}(a, v) \longrightarrow \mathbf{m}(c, @[C[v]])$ for a combinator context C with holes in name positions. This is derivable from the family by the chain of Proposition 3.8, and conversely each member of the family is the instance $C = A(-, \dots, -)$. It shortens the translation and its proofs, at the price of an atom parameterised by a context rather than by names — a considerable conceptual cost, and one we do not think worth paying in the presentation, though it is worth recording that the two formulations have the same reach.

4 Why the original presentation is the right one here

Proposition 3.4 shows the Meredith–Radestock presentation is *necessary* for the combinators. Three further observations show it is fitted to them.

The e rule is the substitution clause. Compare

$$(*y)\{\textcircled{Q}/y\} = Q \quad \text{with} \quad \text{e}(a) \mid \text{m}(a, \textcircled{Q}) \longrightarrow Q.$$

These are the same equation. In the rho calculus the decoding happens at substitution time, in the metatheory; in the combinators, which have no substitution, it happens at communication time, as a reduction. A presentation that instead identifies $*\textcircled{Q}$ with Q by congruence leaves the combinator rule with nothing to correspond to, and must put the decoding into $\equiv$ — where, by Proposition 3.4, it is fatal. The reason is structural rather than notational: the combinators pass names and do not inspect them, so the only way they can reach the process a name encodes is by an explicit step, and an explicit step is a reduction.

The surviving half is load-bearing. $\textcircled{*}x \equiv_{\mathcal{N}} x$ is what makes it possible to send a name at all. A message carries a process, so “send the name x on v ” is $v!(*x)$, with payload $\textcircled{*}x$; the encoding $\text{m}(v, x)$ is legitimate only because $\textcircled{*}x \equiv_{\mathcal{N}} x$.

Correctness lives at $\approx$ in any case. Prefix elimination introduces administrative forwarders (§7), so the image of a source term is never structurally identical to what one would write by hand, only bisimilar to it. A presentation whose decoder may return a bisimilar rather than a congruent process is already calibrated to the theory in which the translation can be stated, and the congruence one wants is recovered downstream, by the equivalence of §5, rather than asserted upfront in $\equiv$.

5 Context-labelled bisimilarity

We use the equivalence of the companion note [4] rather than barbs, for consistency with it. The two are close here — both calculi are name-passing, so barbs are an adequate observation — but the context-labelled form states the restriction on observers as part of the relation rather than as a side condition on a set of names, and that restriction is the substance of Theorem 7.9.

Definition 5.1. For a calculus with reduction $\longrightarrow$ and contexts $\mathcal{C}$, write $P \xrightarrow{\mathcal{C}} P'$ when $C[P] \longrightarrow P'$ and $P \xRightarrow{\mathcal{C}} P'$ when $C[P] \longrightarrow^* P'$. For $\mathcal{D} \subseteq \mathcal{C}$, a symmetric $\mathcal{R}$ is an $\mathcal{D}$ -bisimulation if $P \mathcal{R} Q$ and $P \xrightarrow{\mathcal{C}} P'$ with $C \in \mathcal{D}$ imply $Q \xRightarrow{\mathcal{C}} Q'$ with $P' \mathcal{R} Q'$. Write $\approx_{\mathcal{D}}$ for the largest such.

On the source, $\mathcal{D} = \mathcal{C}_{\text{rho}}$, written $\approx_{\text{rho}}$. On the target, $\mathcal{D} = \llbracket \mathcal{C}_{\text{rho}} \rrbracket$, the encodings of source contexts. As in [4] the labels are enabling contexts, not minimal ones; minimality is the right long-run discipline and is developed in [1], and nothing below needs it. Without it the label set is larger and $\approx_{\mathcal{D}}$ correspondingly finer, which makes completeness harder rather than easier.

6 The translation

6.1 Seeds

The translation is indexed by a name, as the encoding of [4] is indexed by an address, so that it is compositional on the nose.

Definition 6.1. A *seed* for P is a name $s \notin \mathcal{N}^*(P)$ modulo $\equiv_{\mathcal{N}}$. Set $s^\varepsilon = s$, $s^{w0} = @m(s^w, @0)$ and $s^{w1} = @k(s^w)$ for a binary string w .

Lemma 6.2. $s := @P$ is a seed for P ; and if s is a seed for P then the s^w are pairwise distinct modulo $\equiv_{\mathcal{N}}$ and none lies in $\mathcal{N}^*(P)$.

Proof. Work with $\equiv_{\mathcal{N}}$ -normal names, using that $@*x \rightarrow x$ is terminating and confluent. Every name occurring in P is the quotation of a proper subterm, hence smaller than $@P$, so $@P \notin \mathcal{N}^*(P)$. The two constructors are injective on normal forms with disjoint images, so $w \mapsto s^w$ is injective; and each s^w mentions s hereditarily. $\square$

Names written p_i, b, c_i below are s^w for distinct w allocated by position in the source term. Nothing depends on the particular constructors.

Remark 6.3. [5] instead builds each constructed name by quoting the subterm it must be fresh for, so every constructed name carries a copy of the body and the output is quadratic in $|P|$ before any other cost is counted. Seeds give names of size $|s| + O(|w|)$ with the same guarantee.

6.2 Gate and distributor

Definition 6.4. For b, c fresh, $G(p, S) := s(p, b, c) \mid q(b, S) \mid e(c)$; and $D(x; p_0) := 0$ with $p_0 := x$, while for $k \geq 1$

$$D(x; p_0, \dots, p_k) := d(x, p_0, q_1) \mid d(q_1, p_1, q_2) \mid \dots \mid d(q_{k-1}, p_{k-1}, p_k).$$

Lemma 6.5. $G(p, S)$ has no reduction and is not a message. For any v ,

$$G(p, S) \mid m(p, v) \longrightarrow fw(b, c) \mid q(b, S) \mid e(c) \longrightarrow m(c, @S) \mid e(c) \longrightarrow S,$$

and these are the only reductions available. $D(x; p_0, \dots, p_k) \mid m(x, v) \xrightarrow{*} m(p_0, v) \mid \dots \mid m(p_k, v)$ in k steps.

Proof. The three names p, b, c are distinct, so no other pairing is possible, and s discards the payload. Before the forwarder is released there is no fw at b , so $q(b, S)$ is inert by Remark 3.6. The claim for D is immediate. $\square$

The gate is the whole of the difference from Yoshida. It delays a continuation of any size for three atoms and one quotation, because the continuation travels as data.

6.3 Clauses

Occurrences of a bound name y in a body P fall into five kinds. Four are occurrences at which y is used as a name; the fifth is an occurrence inside a quotation, at which a new name must be built.

Definition 6.6 (The translation). On names, $\llbracket @P \rrbracket := @\llbracket P \rrbracket_{s^w}$ where w is the position of that quotation in the enclosing term. On processes,

$$\begin{aligned} \llbracket 0 \rrbracket_s &= 0 \\ \llbracket P \mid Q \rrbracket_s &= \llbracket P \rrbracket_{s^0} \mid \llbracket Q \rrbracket_{s^1} \\ \llbracket x!(P) \rrbracket_s &= m(\llbracket x \rrbracket, @\llbracket P \rrbracket_{s^0}) \\ \llbracket \text{for}(y \leftarrow x)P \rrbracket_s &= D(\llbracket x \rrbracket; p_0, \dots, p_k) \mid G(p_0, B) \mid L_1 \mid \dots \mid L_k \end{aligned}$$

	occurrence of y	in the quoted body	link
(o1)	input subject $\text{for}(z \leftarrow y)S$	subject $\rightarrow c_i$	$\mathbf{b}_l(p_i, c_i)$
(o2)	output subject $y!(S)$	subject $\rightarrow c_i$	$\mathbf{b}_r(p_i, c_i)$
(o3)	forwarded name $v!(*y)$	atom deleted	$\mathbf{fw}(p_i, \llbracket v \rrbracket)$
(o4)	drop $*y$ in process position	$\rightarrow \mathbf{e}(c_i)$	$\mathbf{fw}(p_i, c_i)$
(o5)	inside a quotation $@S, y \in \text{fn}(S)$	subject $\rightarrow c_i$	constructor chain for S

Table 1: Occurrence kinds. In (o5) the link is the chain of constructors of Proposition 3.8 that rebuilds $@[S]$ from the received name, delivering it at c_i , where the body uses it as it would any other name. When several kinds coincide in one atom, the atom is deleted from the body and the links are emitted together.

where $o_1, \dots, o_k$ enumerate the free occurrences of y in P , each o_i has a distinct proxy c_i , L_i and the residue in B are given by Table 1, and B is $\llbracket P \rrbracket_{s_0}$ with every occurrence treated accordingly. $\llbracket - \rrbracket$ extends to contexts with the hole expecting a seed; plugging is instantiation of the seed.

The incoming message is duplicated $k + 1$ times. One copy trips the gate, which releases the continuation with each occurrence of the bound name standing at a proxy. The other k arrive at the links, each converting the received name into the wiring its occurrence needs: $\mathbf{b}_r$ installs a forwarder from the proxy to the received name, which is what an output subject requires; $\mathbf{b}_l$ installs one in the opposite direction, for an input subject; $\mathbf{fw}$ delivers the name, for a forwarded name and for a drop, the latter discharged by the $\mathbf{e}$ in the body; and $\mathbf{cons}$ builds the name the source would have built by substituting under the quotation.

Remark 6.7. Three special cases are cheaper and may be used directly: $\llbracket \text{for}(y \leftarrow x)\mathbf{0} \rrbracket = \mathbf{k}(\llbracket x \rrbracket)$, $\llbracket \text{for}(y \leftarrow x)*y \rrbracket = \mathbf{e}(\llbracket x \rrbracket)$, and $\llbracket \text{for}(y \leftarrow x)v!(*y) \rrbracket = \mathbf{fw}(\llbracket x \rrbracket, \llbracket v \rrbracket)$ for $v \neq y$.

Example 6.8. $P = \text{for}(y \leftarrow a)y!(\mathbf{0})$ has one occurrence, of kind (o2), so $\llbracket P \rrbracket_s = \mathbf{d}(a, p_0, p_1) \mid G(p_0, \mathbf{m}(c, @0)) \mid \mathbf{b}_r(p_1, c)$. Supplying $\mathbf{m}(a, @[R])$ gives, in order, $\mathbf{m}(p_0, @[R]) \mid \mathbf{m}(p_1, @[R])$; then $\mathbf{m}(c, @0) \mid \mathbf{fw}(c, @[R])$; then $\mathbf{m}(@[R], @0) = \llbracket @R!(\mathbf{0}) \rrbracket$.

Example 6.9 (The witness). $W = \text{for}(y \leftarrow a)@(*y \mid *y)!(\mathbf{0})$ has one occurrence, of kind (o5): both drops resolve to the received name, and the node above them is a parallel composition. So

$$\llbracket W \rrbracket_s = \mathbf{d}(a, p_0, p_1) \mid G(p_0, \mathbf{m}(c, @0)) \mid \mathbf{d}(p_1, r_1, r_2) \mid \mathbf{cons}_l(r_1, r_2, c),$$

the inner duplicator supplying the received name to both arguments. Receiving $\mathbf{m}(a, @[R])$ builds $\mathbf{m}(c, @([R] \mid [R]))$, and the released body sends $\mathbf{0}$ on that name, which is $\llbracket @([R] \mid [R])!(\mathbf{0}) \rrbracket$. Theorem 9.2 shows no term of the constructor-free calculus does this.

7 Correctness

Lemma 7.1 (Compositionality). $\llbracket C[P] \rrbracket = \llbracket C \rrbracket[\llbracket P \rrbracket]$, on the nose.

Proof. Induction on C . Each clause of Definition 6.6 mentions its sub-translations only under a seed abstraction, and the hole occurs in exactly one. $\square$

Lemma 7.2 (Linearity). In $\llbracket P \rrbracket_s$ every constructed name is the subject of at most one atom and carries at most one message, and every atom participates in at most one reduction.

Proof. Names are allocated by position, so distinct positions get distinct names by Lemma 6.2; each clause emits each constructed name in one subject position and at most one value position; every rule consumes both participants. Copies arise only as payloads, and a payload carries its own allocation. $\square$

Definition 7.3 (Link expansion). $T' \succeq T$ if T' is obtained from T by replacing the subject a of finitely many atoms by distinct fresh proxies c and adjoining $\text{fw}(c, a)$ for a value-producing subject, $\text{fw}(a, c)$ for a value-consuming one.

Lemma 7.4. *If $T' \succeq T$ and T satisfies Lemma 7.2 then $T' \approx_{\llbracket \mathcal{C}_{\text{rho}} \rrbracket} T$.*

Proof. Relate T' to T when T' is such an expansion with some adjoined forwarders already discharged. A labelled step of T through an expanded atom is matched in T' by the relay step followed by the same rule, giving an expansion of the residue; a step of T' is either a relay, matched by the empty path, or an ordinary step matched directly. Linearity guarantees one one-shot forwarder per occurrence suffices, the atom firing at most once. No adjoined forwarder is a message, so no label distinguishes them. $\square$

Lemma 7.5 (Inertness). $\llbracket \text{for}(y \leftarrow x)P \rrbracket_s$ has no reduction and contains no message.

Proof. The atoms present are the duplicators of D , the gate, and the links. None is a message, and the atoms of B are stored inside $\mathbf{q}$, hence are not subterms of any redex. No forwarder is present, so $\mathbf{q}(b, B)$ is inert. $\square$

Lemma 7.6 (Name usage). *Let y be bound in $\llbracket P \rrbracket_s$ by the translation of a rho input. Every occurrence of the corresponding proxy is the subject of an atom of $\llbracket P \rrbracket_s$ or an argument of a link. In particular an encoded context neither names the constructed apparatus s^w of a term it is composed with, nor installs a forwarder at such a name.*

Proof. Induction on P using Definition 6.6: proxies are introduced only by Table 1 and used only there. For the second claim, $\llbracket C \rrbracket$ allocates its constructed names from its own seed, and by Lemma 6.2 those are distinct from the seed powers of the term plugged into the hole. $\square$

Lemma 7.7 (Release). $\llbracket \text{for}(y \leftarrow x)P \rrbracket_s \mid \mathbf{m}(\llbracket x \rrbracket, @\llbracket R \rrbracket) \longrightarrow^* T$ with $T \approx_{\llbracket \mathcal{C}_{\text{rho}} \rrbracket} \llbracket P\{ @R/y \} \rrbracket_s$, in $k + 3$ steps followed by at most k administrative steps, and no other sequence is available up to the order of independent steps.

Proof. By Lemma 6.5 the distributor delivers a copy to each p_i and the gate releases B . Each remaining copy meets its link. For (o1) and (o2) the link yields a one-shot forwarder, which is the corresponding link expansion; for (o3) it yields the message the translated output would have been; for (o4) it yields $\mathbf{m}(c_i, @\llbracket R \rrbracket)$, which with the $\mathbf{e}(c_i)$ in B gives $\llbracket R \rrbracket$, and $(*y)\{ @R/y \} = R$ by Definition 2.3, so the occurrence is discharged exactly; for (o5) the chain of Proposition 3.8 yields $\mathbf{m}(c_i, @\llbracket S\{ @R/y \} \rrbracket)$, node by node, since $\llbracket - \rrbracket$ is a homomorphism and each constructor reproduces one node of $\llbracket S \rrbracket$ with the received name in place of the occurrence. Hence $T \succeq \llbracket P\{ @R/y \} \rrbracket_s$ and Lemma 7.4 applies. Uniqueness follows from Lemmas 7.5 and 7.2. $\square$

The (o4) case is where semantic substitution and the $\mathbf{e}$ rule coincide; with syntactic substitution the two sides would differ by $*@\llbracket R \rrbracket$ against $\llbracket R \rrbracket$ and the lemma would need the discarded congruence. The (o5) case is where the constructor earns its place.

Lemma 7.8 (Seed independence). $\llbracket P \rrbracket_s \approx_{\llbracket \mathcal{C}_{\text{rho}} \rrbracket} \llbracket P \rrbracket_t$ for seeds s, t of P .

Proof. Let $\beta_{s \rightarrow t}$ send $s^w \mapsto t^w$ and fix all other names; it is a bijection by Lemma 6.2 and extends to terms by metalevel replacement, descending into quotations. It is not a rho substitution and no combinator context performs it. Take $\mathcal{R} = \{(T, \beta_{s \rightarrow t}(T))\}$ for T reachable from $\llbracket P \rrbracket_s$. A bijection on names preserves and reflects $\equiv_{\mathcal{N}}$, hence the applicability of every rule of Definition 3.3, including **e** and **cons**, whose right-hand sides are computed from the redex and so commute with β . For the labels, Lemma 7.6 shows an encoded context never names a seed power of the term it observes, so no label distinguishes T from its image. $\square$

Theorem 7.9 (Full abstraction relative to encoded contexts). *For all rho processes P, Q ,*

$$P \approx_{\mathcal{C}_{\text{rho}}} Q \iff \llbracket P \rrbracket \approx_{\llbracket \mathcal{C}_{\text{rho}} \rrbracket} \llbracket Q \rrbracket.$$

Proof. Forward. Let $\mathcal{S}$ relate T to U whenever T and U are reachable from $\llbracket P' \rrbracket$ and $\llbracket Q' \rrbracket$ respectively, for some $P' \approx Q'$, by the sequences of Lemma 7.7, modulo link expansion and β . It is symmetric and contains $(\llbracket P \rrbracket, \llbracket Q \rrbracket)$. Given a label $\llbracket C \rrbracket$, Lemma 7.1 gives $\llbracket C \rrbracket[\llbracket P' \rrbracket] = \llbracket C[P'] \rrbracket$, so a target step from the plugged term is, by Lemma 7.5, either administrative — matched by the empty path, staying in $\mathcal{S}$ by Lemmas 7.4 and 7.8 — or the first step of the sequence of Lemma 7.7, in which case $C[P'] \rightarrow P''$ in the source and the matching source step for Q' transfers back along the same lemma.

Backward. A source step $C[P] \rightarrow P''$ is, by Lemma 7.7 read left to right, matched by $\llbracket C \rrbracket[\llbracket P \rrbracket] \rightarrow^* T \approx \llbracket P'' \rrbracket$; so a distinguishing source context yields a distinguishing encoded context, and the contrapositive is the claim. $\square$

Proposition 7.10 (The restriction is forced). *There are rho processes $P \approx_{\mathcal{C}_{\text{rho}}} Q$ and a combinator context $C \notin \llbracket \mathcal{C}_{\text{rho}} \rrbracket$ with $C[\llbracket P \rrbracket] \not\approx C[\llbracket Q \rrbracket]$.*

Proof. The seed is computed by the translation from the source, so an arbitrary combinator context can name s^w ; take $C = [\cdot] \mid \text{fw}(b, u)$ for the gate name b of a translated input and an observable u . By Lemma 6.5 this extracts the stored continuation at u , distinguishing two source terms that differ only in a continuation not otherwise reachable. By Lemma 7.6 no encoded context installs such a forwarder. $\square$

Remark 7.11. This is the same shape as the corresponding result in [4], and for the same underlying reason: the target manufactures names the source does not have, and an unrestricted observer of the target may use them. There the names are addresses; here they are seed powers. In neither case is the restriction a weakening of the theorem — it is the theorem’s shape. Note also that **q** is doing real work in Proposition 7.10: with the continuation stored as $\text{m}(b, @B)$ the context would not even need to act, since the store would be a message and hence directly observable.

Remark 7.12 (Loud and quiet). With the constructors in the calculus the point generalises. A construction chain parks each intermediate result as a message on a constructed name, so the whole dataflow of name-building is observable to an unrestricted context, and Table 1 (o5) makes such chains common. The question **q** answers for the nullary case — should a stored value be loud or quiet? — therefore applies at every arity, and a uniform treatment would give each constructor a quiet variant emitting $\text{q}(c, \cdot)$ rather than $\text{m}(c, \cdot)$. We have not done this, and it is the main design question left open by §10.

8 Complexity of the target term

Let $|P|$ count atoms and $d(P)$ the nesting depth of input prefixes.

Proposition 8.1. $|\llbracket P \rrbracket_s| = O(|P|)$, and the textual size is $O(|P| \cdot (|s| + \log |P|))$.

Proof. The clauses for $\mathbf{0}$, parallel composition and output add at most one atom each. An input adds k duplicators, three gate atoms and k links, where k counts occurrences of the bound name; the body is translated once and appears once, inside a quotation. Summing k over binders counts each occurrence once. A link of kind (o5) is a chain of $O(|S|)$ constructors by Proposition 3.8, which is within the same bound since S is a subterm. Textual size follows from $|s^w| = |s| + O(|w|)$ with $O(|P|)$ names allocated. $\square$

Yoshida must distribute the prefix through the body: her rule (I) splits at every parallel composition, inserting a duplicator, and her rules (V) and (VI) delay every atom behind its own synchroniser. Each prefix therefore multiplies the body.

Proposition 8.2. Let $P_d = \text{for}(x_1 \leftarrow a)\text{for}(x_2 \leftarrow a) \cdots \text{for}(x_d \leftarrow a)\mathbf{0}$, so $|P_d| = d + 1$ and $d(P_d) = d$. Yoshida's elimination yields $T(d) = (3^{d-1} + 1)/2$ atoms, where $T(1) = 1$ and $T(j + 1) = 3T(j) - 1$, while $|\llbracket P_d \rrbracket| = 3d + 1$.

Proof. The innermost prefix gives one atom by her rule (III). If the body of the next prefix has n atoms, none mentioning the bound name, rule (I) inserts $n - 1$ duplicators along a binary tree and rule (VI) replaces each atom by a synchroniser and a relabelled copy, giving $2n + (n - 1)$. No other rule applies, so the count is exact. For the reflective encoding each prefix has no occurrences, contributing an empty distributor and one gate. $\square$

Corollary 8.3. Prefix elimination with reflection is exponentially more succinct than without it, in the nesting depth of inputs.

Remark 8.4. The mechanism is that Yoshida cannot store a continuation, so “wait, then behave like T ” must be implemented by rewriting T itself, and a constant-factor expansion composes multiplicatively down a nesting chain. Reflection stores T as the name $@T$ and pays only for the gate. We do not claim 3^d is optimal for her calculus, only that it is what her rules give; an unconditional lower bound would be a statement about all encodings.

9 What reflection needs

9.1 The combinators cannot compute names

Write rhocomb^- for the calculus of Definition 3.3 without the constructors — that is, the published calculus with the defect of Proposition 3.4 repaired.

Lemma 9.1 (No name creation). *If $T \rightarrow T'$ in rhocomb^- then $\mathcal{N}^*(T') \subseteq \mathcal{N}^*(T)$ modulo $\equiv_{\mathcal{N}}$.*

Proof. Every rule of the constructor-free calculus but $\mathbf{e}$ and $\mathbf{q}$ has, on its right, only names occurring on its left. For $\mathbf{e}(a) \mid \mathbf{m}(a, @P) \rightarrow P$ the names of P are hereditarily among those of $@P$; for $\mathbf{fw}(a, b) \mid \mathbf{q}(a, P) \rightarrow \mathbf{m}(b, @P)$ both b and $@P$ occur in the redex. Structural congruence introduces no names. $\square$

The rho calculus has no such property, and that is the expressive content of substitution under a quotation.

Theorem 9.2 (Separation). *Let $W = \text{for}(y \leftarrow a)@(*y \mid *y)!(\mathbf{0})$. No term of rhocomb^- is $\mathcal{D}$ -bisimilar to $\llbracket W \rrbracket$ for any $\mathcal{D}$ containing the contexts $[\cdot] \mid \mathbf{m}(a, @[\![R]\!]) \mid \mathbf{d}(@([\![R]\!] \mid [\![R]\!]), u, u')$.*

Proof. In the source, $W \mid a!(R) \longrightarrow @(\mathbf{0} \mid R)!(\mathbf{0})$ by Definition 2.3, so the corresponding context yields a step at u . Let T be any term of rhocomb^- . $\mathcal{N}^*(T)$ is finite, so choose R with $@([\![R]\!] \mid [\![R]\!]) \notin \mathcal{N}^*(T) \cup \mathcal{N}^*(@[\![R]\!])$. By Lemma 9.1 no reduct of $T \mid \mathbf{m}(a, @[\![R]\!])$ has that name, so no message is ever sent on it, the context's duplicator never fires, and the label is not matched. $\square$

Remark 9.3. The theorem is not about quoting a parallel composition in particular: any source term whose subject is a quotation mentioning a bound name has the same character. The slogan is that rhocomb^- can quote code but cannot compute names. Example 6.9 shows $\text{cons}_\downarrow$ supplies exactly the missing step, so the pair of results is a characterisation rather than an obstruction: the combinators fall short of the rho calculus by one operation, and here it is.

Remark 9.4. There is an instructive mirror image in [2], used in [4]: the rho calculus manufactures new free — hence observable and substitutable — names at runtime, which is why no encoding of the π -calculus into rho is fully abstract against unrestricted rho observers. The same capability read from the other side is why the constructor-free combinators cannot encode rho at all. One calculus has too much name construction for π to be safe in it; the other has too little for rho to fit.

9.2 Minimality

Proposition 9.5. *Let the calculus be taken without $\mathbf{e}$. Then every reduction sequence from T has length at most the number of non-message atoms of T ; so it is strongly normalising, hence not Turing complete, and $\mathbf{e}$ is not definable in it.*

Proof. Each rule consumes one non-message atom and produces none, except that $\mathbf{b}_r$, $\mathbf{b}_l$ and $\mathbf{s}$ produce fw while consuming themselves. The rule for $\mathbf{q}$ consumes two non-message atoms and produces a message; each constructor rule consumes one non-message atom and its input messages, and produces one message. So the count strictly decreases at every step. The rule for $\mathbf{e}$ is the only one whose right-hand side is an arbitrary process. $\square$

So the drop combinator carries the computational power that reflection adds, and the recursion construction of §6 depends on it. This is the analogue of Yoshida's minimality arguments, with a shorter proof because the invariant is an atom count. Note that the constructors do not restore Turing completeness on their own: they build names, and without $\mathbf{e}$ no name is ever opened.

10 An alternative presentation: constructors only

The calculus of §3 has three kinds of atom doing three jobs: routers ($\mathbf{d}, \mathbf{k}, \text{fw}, \mathbf{b}_l, \mathbf{b}_r, \mathbf{s}$), which move names; a destructor ($\mathbf{e}$), which opens one; and value-bearing atoms ($\mathbf{m}, \mathbf{q}, \text{cons}$). The third group is not three things but one.

Remark 10.1. Read a constructor as a dataflow node: $\text{cons}_A(a_1, \dots, a_n, c)$ says that when names are available at $a_1, \dots, a_n$, the name A of them is available at c . Then $\mathbf{m}(c, v)$ is the nullary case — the name v is available at c , unconditionally — and $\mathbf{q}(c, v)$ is the nullary case in a quiet variant. On this reading neither $\mathbf{m}$ nor $\mathbf{q}$ is primitive: the calculus consists of routers, one destructor, and a constructor family whose arity-zero members are the messages.

Two things recommend this presentation. It is uniform: every atom takes names and every rule is either routing, opening, or building, with no atom indexed by a process. And it makes Remark 7.12 a question about the family rather than about two special atoms — whether a constructor’s output is loud or quiet is a single design parameter, and $\mathbf{m}$ versus $\mathbf{q}$ is its nullary instance.

Two things count against it, and we record them rather than settle them. Barbs, or their context-labelled counterpart, need a notion of message; if messages are nullary constructors then observation must be defined on the constructor family, which is a mild complication but changes what Lemma 7.6 has to say. And a calculus closed under its own constructions would need one member per atom shape, including members for the constructors themselves, since a constructed name may quote a term containing one; the four of Proposition 3.8 suffice for the image of the translation but not for that closure. The context-indexed atom of Remark 3.10 sidesteps the regress at a conceptual price we judge too high. Which to prefer is a genuine choice, and the results of this note hold either way.

11 Corrigenda to [5]

Restricted to what bears on the rho calculus; the material specific to the π -calculus route is out of scope here.

1. **Structural congruence.** Delete $\ast(@P) \equiv P$ (Proposition 3.4), and move the drop combinator from the process production to the atom production (Remark 3.5).
2. **No rho-to-combinator translation is given.** The outline promises that the encoding of a source calculus into the combinators composes with an encoding into the rho calculus, but the translation defined in the body has Yoshida terms as its domain. §6 supplies the missing map, and §9 shows that no such map exists without a name constructor.
3. **No prefix-elimination lemma.** The draft states two theorems and proves no lemmas. The prefixed form $\text{for}(x \leftarrow p)C$ is never given a semantics in the combinator calculus, so the correctness of elimination is neither stated nor proved. Lemmas 6.5 to 7.7 are the analogue here.
4. $\equiv_{\mathcal{N}}$ **is undefined for the combinators** yet is used in the side condition on observation.
5. **Name sizes.** Constructed names quote the subterm they must be fresh for, making the output quadratic before other costs (Remark 6.3).
6. **Typing slips.** $x^l := @b_l(x, \mathbf{m}(x, \ast x))$ puts a process in the second argument of b_l , which takes names. The derivation following the definition of the replication server carries a spurious extra copy of $\mathbf{m}(x, @(D \mid P))$ on its third line; the calculation is otherwise correct. Polarity annotations appear inside quoted terms, and several where-clauses construct names mentioning a bound variable, which is a metavariable rather than a name of the calculus.

12 Open questions

1. **Minimal labels.** Theorem 7.9 uses enabling contexts. The minimal-label form, via relative pushouts taken in the image of the translation, is the discipline of [1]; the forward half descends for free and the backward half is what is owed.
2. **Loud and quiet constructors** (Remark 7.12). Should every constructor have a quiet variant, and does the quiet calculus admit a sharper form of Proposition 7.10?
3. **Name growth.** With the constructors, names are no longer subterms of the program: they grow at runtime. The consequences for the decidability of $\equiv_{\mathcal{N}}$ in practice, and for cost accounting, want examining.
4. **Is four minimal?** Proposition 3.8 gives an upper bound for the uniform translation, and Remark 3.9 shows the count is translation-relative. Is there a translation needing fewer, is $\text{cons}_{\downarrow}$ alone enough for some reformulation, and does reflection make any of Yoshida’s six routers redundant?

References

- [1] C.B. Wells, M. Stay and L.G. Meredith. *Behavior in higher-order languages: admissible observers, relative pushouts, and the encoding of extended theories*. In preparation, F1R3FLY.io, 2026.
- [2] S. Lybech. *Encodability and separation for a reflective higher-order calculus*. EXPRESS/SOS 2022, EPTCS 368, 95–112.
- [3] L.G. Meredith and M. Radestock. *A reflective higher-order calculus*. Electronic Notes in Theoretical Computer Science 141(5):49–67, 2005.
- [4] L.G. Meredith. *Namespaces without a name server: freshness by communication, and a fully abstract encoding of the π -calculus in the rho calculus*. F1R3FLY.io, 2026.
- [5] L.G. Meredith and M. Stay. *Name-free combinators for concurrency*. Working draft, rho4u.
- [6] N. Yoshida. *Minimality and separation results on asynchronous mobile processes: representability theorems by concurrent combinators*. Theoretical Computer Science 274(1–2):231–276, 2002.